

Heaviside Composite Optimization Problems

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Chapter I

Introduction

A bird's eye-view of post World-War-II optimization till now:

- From linear programming to nonlinear and integer programming.
- From nonlinear programming to stochastic programming and modern-day convex programming.
- From differentiability to nondifferentiability.
- From nondifferentiability to semicontinuity.
- From convexity to nonconvexity.

Y. Cui and J. S. Pang, *Modern Nonconvex Nondifferentiable Optimization*. SIAM, MOS–SIAM Series on Optimization (December 2021).

Apart from further developments of the present, what is the future??

From semicontinuity to lack thereof $\Rightarrow$ discontinuous optimization.

From closedness of constraints to non-closedness.

The General Heaviside Composite Optimization Problem (G-HSCOP)

$$\begin{aligned} & \underset{\mathbf{x} \in P}{\text{maximize}} \quad \Phi(\mathbf{x}) \triangleq c(\mathbf{x}) + \sum_{k=1}^{K_0} \psi_{0k}(\mathbf{x}) \underbrace{\mathbb{1}_{[0,\infty)}(\phi_{0k}(\mathbf{x}))}_{\text{composite indicator fnc.}} \\ & \text{subject to} \quad \mathbf{A}_{i\bullet} \mathbf{x} + \sum_{k=1}^{K_i} \psi_{ik}(\mathbf{x}) \mathbb{1}_{[0,\infty)}(\phi_{ik}(\mathbf{x})) \geq \eta_i, \quad i = 1, \dots, I. \end{aligned}$$

- P is a **friendly** set, c is a **friendly** function, e.g., affine or quadratic; **friendly** = amenable to well-known treatments, and $\mathbf{A}_{i\bullet}$ is the i -th row of a matrix $\mathbf{A} \in \mathbb{R}^{I \times n}$.
- ψ_{ik} and ϕ_{ik} are functions whose properties highlight the challenges;
- most importantly, $\mathbb{1}_{[0,\infty)}$ is the “closed” Heaviside function given by

$$\mathbb{1}_{[0,\infty)}(s) \triangleq \begin{cases} 1, & \text{if } s \in [0, \infty), \\ 0, & \text{otherwise,} \end{cases} \quad (\text{upper semi-continuous}).$$

The General Heaviside Composite Optimization Problem (G-HSCOP)

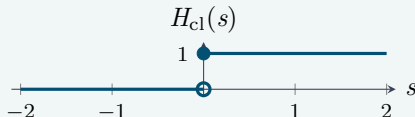
Supplementary visualization: closed vs. open Heaviside thresholds

The Heaviside indicator turns a scalar condition into a binary switch. The bracket at $s = 0$ determines both activation at equality and the direction of semicontinuity.

Closed threshold: $[0, \infty)$

Upper, not lower, semicontinuous; favorable for max.

$$H_{\text{cl}}(s) \triangleq \mathbb{1}_{[0, \infty)}(s) = \begin{cases} 0, & s < 0, \\ 1, & s \geq 0. \end{cases}$$

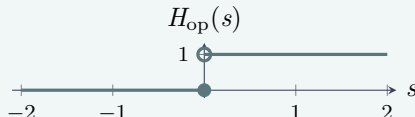


$H_{\text{cl}}(0) = 1$; equality activates the term.

Open threshold: $(0, \infty)$

Lower, not upper, semicontinuous; favorable for min.

$$H_{\text{op}}(s) \triangleq \mathbb{1}_{(0, \infty)}(s) = \begin{cases} 0, & s \leq 0, \\ 1, & s > 0. \end{cases}$$



$H_{\text{op}}(0) = 0$; strict inequality is required. In particular, $|t|_0 = H_{\text{op}}(|t|)$.

Thus $\psi(\mathbf{x})H_{\text{cl}}(\phi(\mathbf{x}))$ equals $\psi(\mathbf{x})$ when $\phi(\mathbf{x}) \geq 0$, and equals 0 otherwise.

Conditional Functions and Sparsity

The product $f(\mathbf{x})\mathbb{1}_{[0,\infty)}(g(\mathbf{x}))$ is a conditional function; i.e., the primary function $f(\mathbf{x})$ is activated only if the auxiliary condition $g(\mathbf{x}) \geq 0$ holds.

An immediate example: the sparsity (counting) function

$$|t|_0 \triangleq \begin{cases} 1, & \text{if } t \neq 0, \\ 0, & \text{if } t = 0, \end{cases} = \mathbb{1}_{(0,\infty)}(|t|) = 1 - \mathbb{1}_{[0,\infty)}(-|t|),$$

illustrates that the function inside the Heaviside function may not be differentiable. This function extends to any step function with a finite number of (discontinuous) jumps.

An **affine sparsity** constraint

$$\sum_{j=1}^n a_j |x_j|_0 \leq b, \quad \text{with } a_j \text{ of mixed signs,}$$

is a non-trivial extension of the cardinality constraint $\sum_{j=1}^n |x_j|_0 \leq K$, whose mixing coefficients are all positive.

Supplementary note. Positive coefficients preserve lower semicontinuity. With mixed-sign a_j , upward and downward jumps can coexist, so $\sum_{j=1}^n a_j |x_j|_0$ may be neither lower nor upper semicontinuous.

References

Methodology

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- S. HAN, Y. CUI, AND J.S. PANG. Analysis of a class of minimization problems lacking lower semicontinuity. *Mathematics of Operations Research* 50(3): 2175–2198 (2025-8).
- K. ZHENG, J. LIU, R. WANG, AND J.S. PANG. Solving constrained affine Heaviside composite optimization problems by a progressive IP approach. Manuscript (April 2026).

Applications

- Z. QI, Y. CUI, Y. LIU, AND J.S. PANG. Estimation of individualized decision rules based on optimized covariate-dependent equivalent of random outcomes. *SIAM Journal on Optimization* 29(3): 2337–2362 (2019).
- Y. FANG, J. LIU, AND J.S. PANG. Treatment learning with Gini constraints by Heaviside composite optimization and a progressive method. *Computational Optimization and Applications* 92(2): 471–513 (2025-11).

Distinguished (and challenging) features:

- **Discontinuity** of the Heaviside function and **nondifferentiability** of the other functions:
 - lack of semicontinuity in the objective function
 - non-closedness of the feasible set
 - does convexity/concavity have a role to play?
- **Composition** of the highlighted features:
 - what is a reasonable concept of a **solution**?
 - algorithmically, how to compute such a solution?
- **Fusion** of continuous variables and functions with discrete structure of the Heaviside function; thus natural to consider
 - a **marriage** between the continuous and discrete world
 - how to facilitate the marriage? will it be fruitful?

Chapter II

Prerequisites, a second course in optimization

Prerequisites, a second course in optimization

- A graduate course in linear, differentiable, and integer programming
- **Bouligand differentiability**: broad and most basic for nondifferentiable functions
- **Stationarity conditions** of nondifferentiable programs
- **Piecewise affine functions**: definition and properties
- **Difference-of-convex programs**: criticality vs. stationarity; the difference-of-convex algorithm and its refinement
- **Complementarity constraints**: modeling breadth and connections to binary programming
- **Fractional and composite programs**: Dinkelbach's algorithm and its extension.

Supplementary note. $0 \leq a \perp b \geq 0$ is an or-gate — $a_i = 0$ or $b_i = 0$ — i.e. if-then logic written in continuous variables.

Bouligand differentiability

A function $f: \mathcal{O} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, where $\mathcal{O}$ is an open set in $\mathbb{R}^n$ is **Bouligand differentiable** at $\bar{\mathbf{x}} \in \mathcal{O}$ if it is **locally Lipschitz continuous** near $\bar{\mathbf{x}}$ and directionally differentiable there with the (finite) directional derivative along the direction $\mathbf{v} \in \mathbb{R}^n$ defined by:

$$f'(\bar{\mathbf{x}}; \mathbf{v}) \triangleq \lim_{\tau \downarrow 0} \frac{f(\bar{\mathbf{x}} + \tau \mathbf{v}) - f(\bar{\mathbf{x}})}{\tau}.$$

As a function of the direction, $f'(\bar{\mathbf{x}}; \bullet)$ is positively homogeneous and globally Lipschitz continuous on $\mathbb{R}^n$ with the same Lipschitz modulus as that of f near $\bar{\mathbf{x}}$.

Calculus rules hold for the directional derivative; these are the sum, multiplication, division, and composition rules; in particular,

$$(f \circ g)'(\bar{\mathbf{x}}; \mathbf{v}) = f'(g(\bar{\mathbf{x}}); g'(\bar{\mathbf{x}}; \mathbf{v})),$$

Remark: the partial differentiation rule may fail;

$$f'(\bar{\mathbf{x}}; \mathbf{v}) \neq \sum_{i=1}^n f'(\bar{\mathbf{x}}; v_i \mathbf{e}^i), \quad \text{where } \mathbf{e}^i = i\text{-th coordinate vector}$$

Supplementary note. Convex functions are directionally differentiable.

Piecewise affine functions

A **continuous** function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **piecewise affine** if there exists a partition of $\mathbb{R}^n$ into a finite number of polyhedra, on each of which, f is an affine function.

A piecewise affine function is Bouligand differentiable everywhere. **Most interested in $m = 1$.**

A **continuous** function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is piecewise affine (PA) if and only if

$$f(\mathbf{x}) = \underbrace{\max_{1 \leq k \leq K} [(\mathbf{a}^k)^\top \mathbf{x} + \alpha_k]}_{\text{convex in } \mathbf{x}} - \underbrace{\max_{1 \leq \ell \leq L} [(\mathbf{b}^\ell)^\top \mathbf{x} + \beta_\ell]}_{\text{convex in } \mathbf{x}}, \quad \mathbf{x} \in \mathbb{R}^n$$

for some constant vectors $\mathbf{a}^k$ and $\mathbf{b}^\ell$ and scalars α_k and β_ℓ . Hence, if f is PA, then for every $\bar{\mathbf{x}} \in \mathbb{R}^n$,

$$f(\mathbf{x}) = f(\bar{\mathbf{x}}) + f'(\bar{\mathbf{x}}; \mathbf{x} - \bar{\mathbf{x}}), \quad \forall \mathbf{x} \text{ sufficiently near } \bar{\mathbf{x}},$$

Chapter III

Paradigm changes: Sources of discontinuity

Modeling breadth (OR, statistics, ML, and policy learning)

- **Set-up costs**— classical in operations research, today's term is sparsity
- **Structured variable/constraint selection**; on-off constraints
- **Conditional stochastic functionals** — probabilities and expectations
- **Statistical models** — conformal regression; change-point detection; robust trimming; and more
- **Classification** — score-based binary classification; multiclass classification; advanced measures; decision trees
- **Clustering: refreshed and extended** — mixed-norm: extending traditional K-means and K-clustering; bicriteria (errors versus similarities)
- **Univariate and vector quantization**
- **Policy learning and causal inference** — piecewise affine policy for stochastic optimization; causal inference for individualized treatment learning; policy learning with optimal weight clipping

A host of instances

from classical to modern to future

Hierarchical variable selection

The statistical model with interaction terms: hypothesizing a bilinear relation among the input features $\boldsymbol{\xi} = (\xi_i)_{i=1}^d$

$$m(\mathbf{x}; \boldsymbol{\xi}) = \sum_{i=1}^d x_i^{(1)} \xi_i + \sum_{1 \leq i < j \leq d} x_{ij}^{(2)} \xi_i \xi_j,$$

with the parameter $\mathbf{x}$ consisting of the unknown model coefficients $x_i^{(1)}$ and $x_{ij}^{(2)}$ that obey

- **strong hierarchy**: an interaction term can be selected only if both of the linear terms are selected, i.e.,

$$|x_{ij}^{(2)}|_0 \leq \min\{|x_i^{(1)}|_0, |x_j^{(1)}|_0\};$$

- **weak hierarchy**: an interaction term can be selected only if at least one of the corresponding linear terms is selected,

$$|x_{ij}^{(2)}|_0 \leq |x_i^{(1)}|_0 + |x_j^{(1)}|_0;$$

both leading to **mixed-sign** combinations of Heaviside functions.

Conditional stochastic functionals

- Expression of a probability function as an expectation:

$$\mathbb{P}[f(\mathbf{x}, \boldsymbol{\xi}) \geq 0] = \mathbb{E} \mathbb{1}_{[0, \infty)}(f(\mathbf{x}, \boldsymbol{\xi})) \xrightarrow[\text{approximation (SAA)}]{\text{sample average}} \frac{1}{N} \sum_{s=1}^N \mathbb{1}_{[0, \infty)}(f(\mathbf{x}, \boldsymbol{\xi}^s));$$

- and by extension, expression of a **decision-dependent conditional expectation function**:

$$\mathbb{E}[\psi(\mathbf{x}, \boldsymbol{\xi}) \mid f(\mathbf{x}, \boldsymbol{\xi}) \geq 0] \triangleq \underbrace{\frac{\mathbb{E}[\psi(\mathbf{x}, \boldsymbol{\xi}) \mathbb{1}_{[0, \infty)}(f(\mathbf{x}, \boldsymbol{\xi}))]}{\mathbb{P}[f(\mathbf{x}, \boldsymbol{\xi}) \geq 0]}}_{\text{fractional Heaviside function}}.$$

Note the product

$$\psi(\mathbf{x}, \boldsymbol{\xi}) \mathbb{1}_{[0, \infty)}(f(\mathbf{x}, \boldsymbol{\xi}))$$

raising the possibility of losing semicontinuity (in the decision variable $\mathbf{x}$) if the sign of the multiplier function $\psi(\bullet, \boldsymbol{\xi})$ is not consistent with the optimization.

Mixed-sign combinations and piecewise inner functions

- First-order stochastic dominance:

$$\mathbb{P}[f_1(\mathbf{x}, \boldsymbol{\xi}) \leq c_1] \leq \mathbb{P}[f_2(\mathbf{x}, \boldsymbol{\xi}) \leq c_2]$$

- Conjunctive probability constraint:

$$\begin{aligned} b &\geq \mathbb{P}[\psi_1(\mathbf{x}, \boldsymbol{\xi}) \geq 0 \text{ and } \psi_2(\mathbf{x}, \boldsymbol{\xi}) < 0] \\ &= \mathbb{P}[\psi_1(\mathbf{x}, \boldsymbol{\xi}) \geq 0] - \mathbb{P}[\min(\psi_1(\mathbf{x}, \boldsymbol{\xi}), \psi_2(\mathbf{x}, \boldsymbol{\xi})) \geq 0] \end{aligned}$$

- Conditional probability constraint:

$$\begin{aligned} b &\geq \mathbb{P}[\psi_1(\mathbf{x}, \boldsymbol{\xi}) \geq 0 \mid \psi_2(\mathbf{x}, \boldsymbol{\xi}) \geq 0] \\ &\quad \Updownarrow \\ \mathbb{P}[\min(\psi_1(\mathbf{x}, \boldsymbol{\xi}), \psi_2(\mathbf{x}, \boldsymbol{\xi})) \geq 0] - b \mathbb{P}[\psi_2(\mathbf{x}, \boldsymbol{\xi}) \geq 0] &\leq 0. \end{aligned}$$

Some statistical models (communicated by Dr. Yao Xie)

- **Change-point detection with sparsity.** Let $\{\mathbf{x}_t\}_{t=1}^T$ be a sequence with $\mathbf{x}_t \in \mathbb{R}^p$ and multiple unknown change point $\{\tau_i\}_{i=1}^I$ with mean shifts $\{\Delta_i\}_{i=1}^I$ to be determined. Let $\mu \in \mathbb{R}^p$ be the baseline mean vector which is unknown. This problem is formulated as follows:

$$\begin{aligned} \underset{\mu, \{\Delta_i\}, \{\tau_i\}}{\text{minimize}} \quad & \sum_{t=1}^T \mathbb{1}_{(0, \infty)}(\tau_1 - t) \|\mathbf{x}_t - \mu\|_2^2 + \sum_{i=1}^I \|\Delta_i\|_0 \\ & + \sum_{i=1}^I \sum_{t=1}^T \underbrace{\mathbb{1}_{(0, \infty)}(t - \tau_i) \mathbb{1}_{[0, \infty)}(\tau_{i+1} - t)}_{\text{expressing } \tau_i < t \leq \tau_{i+1}} \|\mathbf{x}_t - \mu - \Delta_i\|_2^2 \end{aligned}$$

subject to $\tau_1 \leq \tau_2 \leq \dots \leq \tau_I$.

Note the product of an open and a closed Heaviside function.

Supplementary note. Matched brackets let two factors collapse into one, $\mathbb{1}_{[0, \infty)}(a) \mathbb{1}_{[0, \infty)}(b) = \mathbb{1}_{[0, \infty)}(\min(a, b))$, and likewise for $\mathbb{1}_{(0, \infty)}$. The mixed pair above does not, since $\{a > 0, b \geq 0\}$ is neither open nor closed. Under $\tau_i \leq \tau_{i+1}$ it is instead the mixed-sign difference $\mathbb{1}_{[0, \infty)}(\tau_{i+1} - t) - \mathbb{1}_{[0, \infty)}(\tau_i - t)$, with affine inner functions.

- Trimmed least-squares for robust statistical learning.

$$\begin{array}{l} \underset{f \in \mathcal{F}; \gamma \geq 0}{\text{minimize}} \quad \underbrace{\sum_{i=1}^n \mathbb{1}_{[0, \infty)}(\underbrace{\gamma - \|\mathbf{y}_i - f(\mathbf{x}_i)\|}_{\text{trimming threshold}}) \|\mathbf{y}_i - f(\mathbf{x}_i)\|_2^2}_{\text{a conditional error}} \end{array}$$

$$\begin{array}{l} \text{subject to} \quad \underbrace{\frac{1}{n} \sum_{i=1}^n \mathbb{1}_{[0, \infty)}(\gamma - \|\mathbf{y}_i - f(\mathbf{x}_i)\|)}_{\text{bounding the trimming ratio}} \geq 1 - \beta. \end{array}$$

Clustering problems: Classical and extended.

Let $\Xi \triangleq \{\xi^s\}_{s=1}^N$ be a given set of data points in $\mathbb{R}^p$. Let $\mathbf{c} \triangleq \{\mathbf{c}^k\}_{k=1}^K$ be an unknown set of cluster centers in $\mathbb{R}^p$. The criterion for a data point ξ^s to belong to cluster k , written as $s \in k$, is

$$\begin{aligned}\|\xi^s - \mathbf{c}^k\| &\leq \min_{k' \neq k} \|\xi^s - \mathbf{c}^{k'}\| \\ \Leftrightarrow h_s(\mathbf{c}) &\triangleq \min_{k' \neq k} \|\xi^s - \mathbf{c}^{k'}\| - \|\xi^s - \mathbf{c}^k\| \geq 0.\end{aligned}$$

Define the simple indicator

$$z_{sk} \triangleq \mathbb{1}(s \in k) \triangleq \mathbb{1}_{[0, \infty)}\{h_s(\mathbf{c})\} = \begin{cases} 1 & \text{if } s \in k \\ 0 & \text{if } s \notin k \end{cases}$$

The classical K-means ($p = 2$) and K-medians ($p = 1$) problems:

$$\begin{aligned}&\underset{\mathbf{c}; \mathbf{z}}{\text{minimize}} && \frac{1}{N} \sum_{s=1}^N \left[\sum_{k=1}^K z_{sk} \|\xi^s - \mathbf{c}^k\|_p^p \right] \\&\text{subject to} && \sum_{k=1}^K z_{sk} = 1, \quad \forall s = 1, \dots, N && \text{every sample belongs to} \\&&& && \text{at least one cluster} \\&\text{and} && z_{sk} \in \{0, 1\}, \quad \forall (s, k) \in [N] \times [K].\end{aligned}$$

Clustering problems: Classical and extended. (cont.)

Unifying the above two problems is the mixed $\ell_{p,q}$ -clustering problem with arbitrary pairs of integers $p, q \geq 1$:

$$\begin{aligned} & \underset{\mathbf{c}, \mathbf{z}}{\text{minimize}} && \underbrace{\frac{1}{N} \sum_{s=1}^N \left[\sum_{k=1}^K z_{sk} \|\boldsymbol{\xi}^s - \mathbf{c}^k\|_p^p \right]}_{\text{bi-convex; denoted } \text{WCD}_p(\mathbf{c}, \mathbf{z})} && \ell_p\text{-norm in objective} \\ & \text{subject to} && \sum_{k=1}^K z_{sk} = 1, \quad \forall s \in [N] \triangleq \{1, \dots, N\} \\ & \text{and for all } (s, k) \in [N] \times [K]: && \\ & && \underbrace{z_{sk} \leq \mathbb{1}_{[0, \infty)} \left(\min_{k' \neq k} \|\boldsymbol{\xi}^s - \mathbf{c}^{k'}\|_q - \|\boldsymbol{\xi}^s - \mathbf{c}^k\|_q \right)}_{\text{Heaviside constraint with } \ell_q\text{-norm}} \\ & && z_{sk} \in \{0, 1\}, \quad (\text{binary variables}) \end{aligned}$$

Note: The Heaviside constraint is not needed in the classical K-means and K-medians problems where $p = q$.

Supplementary note. When $p = q$ the objective alone decides membership. Writing the indicator out separates the cost (ℓ_p) from the rule (ℓ_q), making the rule a free modeling choice.